

# **Command line seminar-1**

**Bioinformatics course  
Term 2, 2024  
Skoltech**

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**Seminar credits:**

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Asia Mendelevich**

# Outline

- Solving exercises:

Directories, navigation

Full and relative paths

Operations with files and directories

Chaining commands and Find matching word or pattern

- Command line tricks
- Start doing home assignment (finish at class or at home)
- Q&A

**Arrow up/down** – navigation through the history of commands

**Tab** – autofill the path (start typing the path and press Tab for autofill)

## Hot keys:

Ctrl + C – abort

Ctrl + D – exit terminal

## Copy-paste:

- Windows command line / Putty: **Ctrl + Insert** for copying and **Shift + Insert** or **Right mouse button** for pasting.
- Unix: set convenient key combination in terminal parameters.

# Solving exercises



# Full and relative paths

Change to a directory using a relative path:

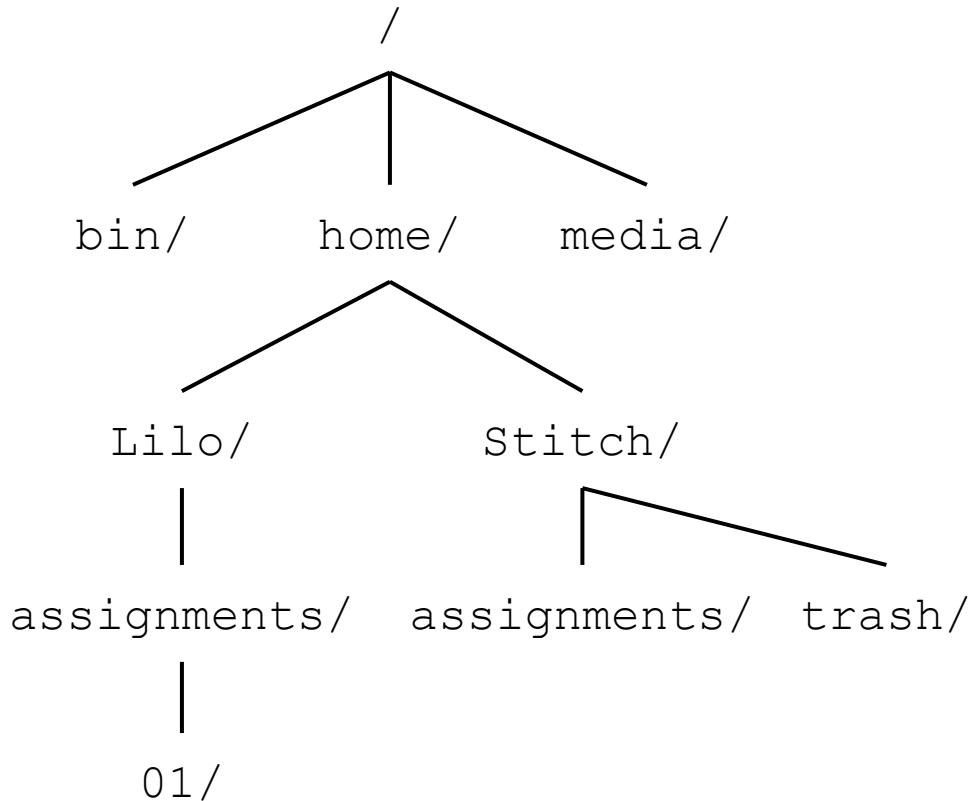
a) from Lilo's home directory to `/media/`

b) from `/home/Stitch/assignments/`  
to `/home/Lilo/assignments/`

```
cd ../../Lilo/assignments/
```

c) from `/home/Lilo/assignments/01/` to  
`/home/Stitch/`

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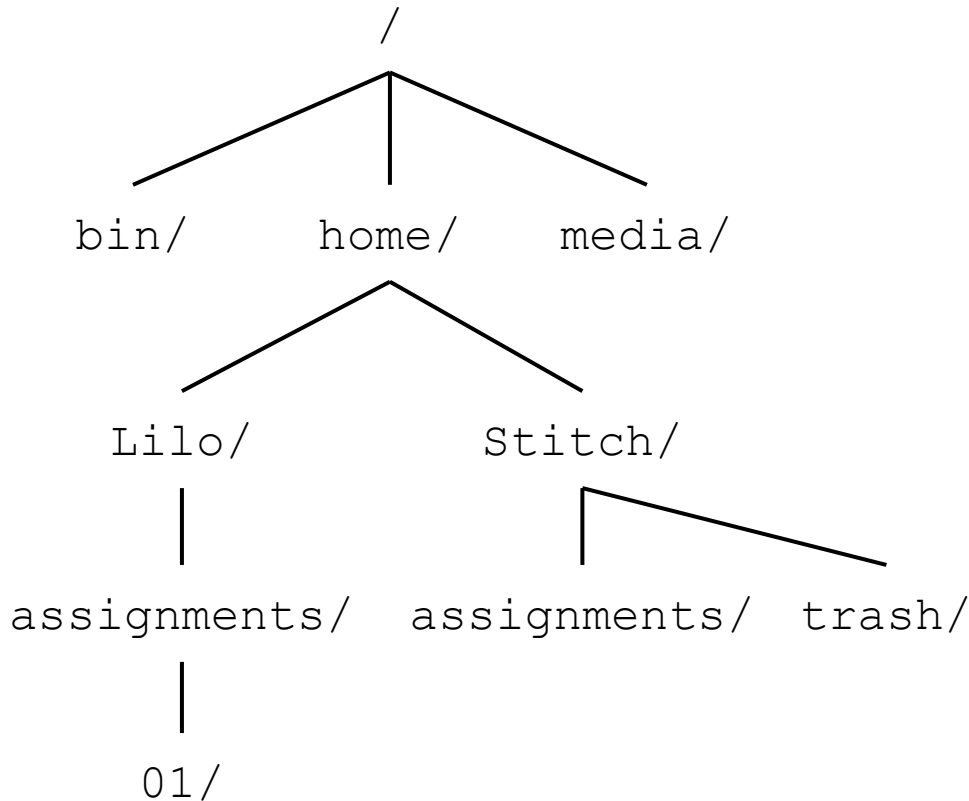
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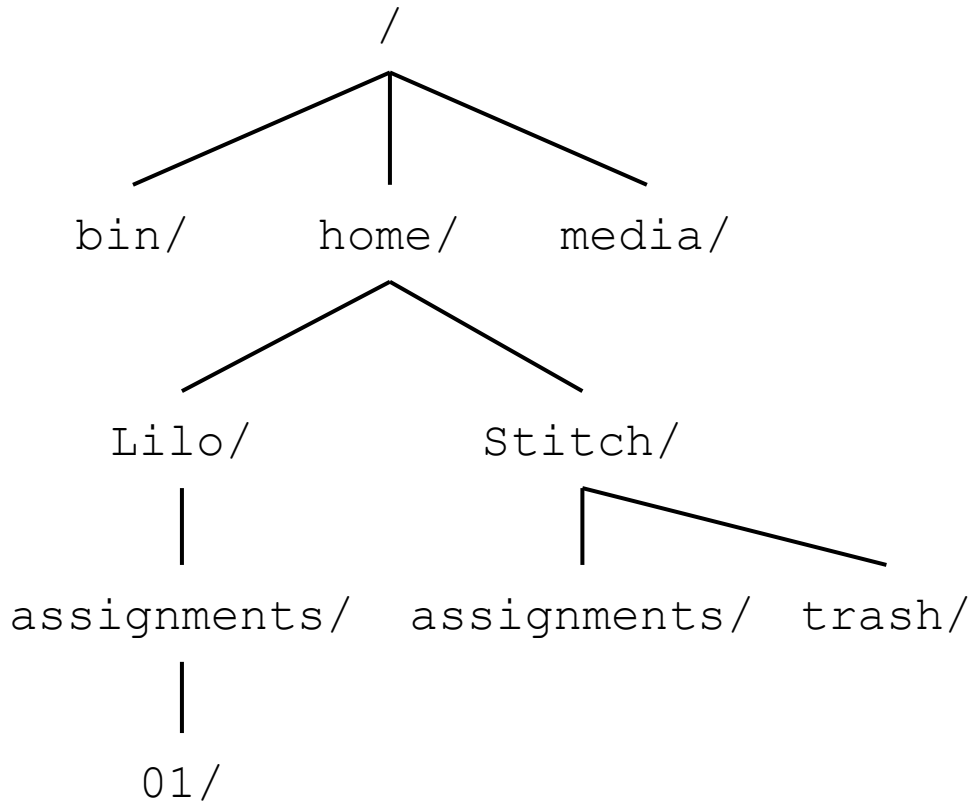
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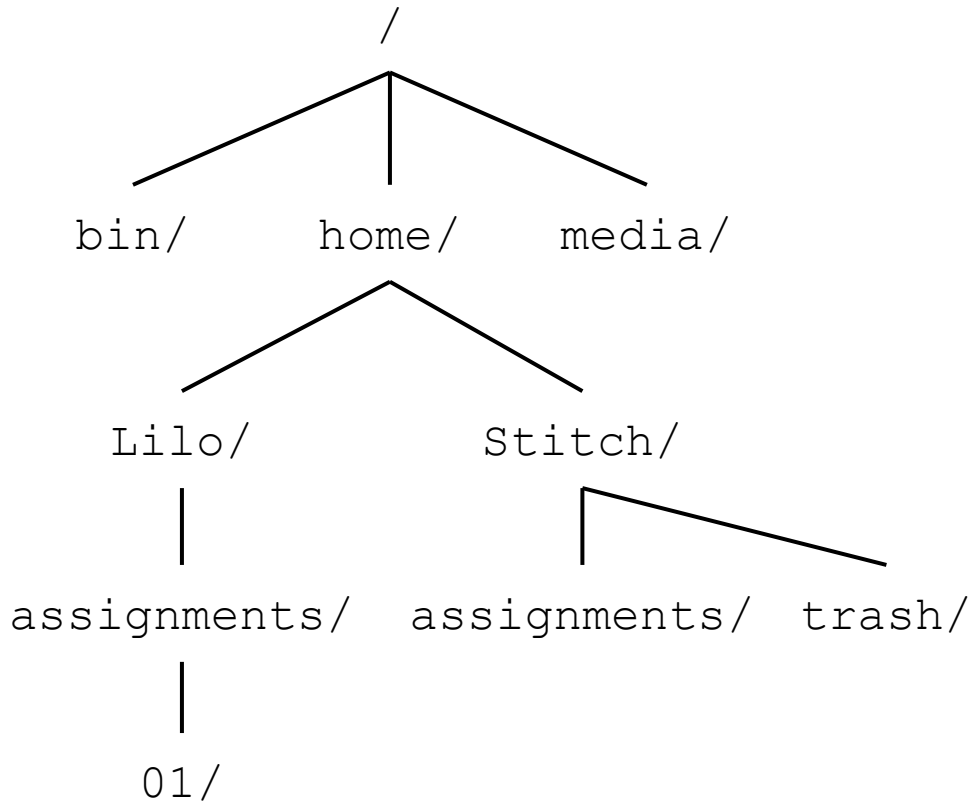
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# Operations with files, text search

Download a file by the link and answer the questions:

[https://files.wwpdb.org/pub/pdb/derived\\_data/pdb\\_entry\\_type.txt](https://files.wwpdb.org/pub/pdb/derived_data/pdb_entry_type.txt)

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head, less, more – cat is not appropriate for long files.

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- d) [tricky] How many structures of nucleic acids are obtained by NMR?

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`grep nuc pdb_entry_type.txt | grep -v prot | grep NMR -c`

# Operations with files

Create a file `readme.md` with the contents below using as many ways as possible:

```
# Documentation to my project

Some description
```

P.S. Markdown is widely used to format documentation on resources like Github as well as the text that accompanies code in, for example, Jupyter Notebooks.

<https://markdownlivepreview.com/>

<https://docs.github.com/ru/get-started/writing-on-github/getting-started-with-writing-and-formatting-on-github/basic-writing-and-formatting-syntax>

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What commands will change the contents of the file `readme.md`?

- a) `tac readme.md`
- b) `cat readme.md > readme.md`
- c) `touch readme.md`
- d) `zip project.zip readme.md`
- e) `head readme.md`
- f) `cp readme.md README.MD`
- g) `mv readme.md README.MD`
- h) `echo '# Documentation to my project\n\nSome description' > readme.md`
- i) `echo '# Documentation to my project\n\nSome description' >> readme.md`
- j) `echo -e '# Documentation to my project\n\nSome description' > readme.md`
- k) `echo -e '# Documentation to my project\n\nSome description' >> readme.md`
- l) `echo '# Documentation to my project'; echo; echo; echo 'Some description' > readme.md`

# Filename expansion (globbing)



**Filename expansion (globbing)** is expanding filename **patterns** with **special characters** (behaviour is different from regex).

**\*** – wildcard for zero or more characters

<https://tldp.org/LDP/abs/html/globbingref.html>

[https://se.ifmo.ru/~ad/Documentation/Shells\\_by\\_Example/ch13lev1sec9.html](https://se.ifmo.ru/~ad/Documentation/Shells_by_Example/ch13lev1sec9.html)

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Examples:

**ls \*.txt** – list all files ending with **.txt**

**cp ????.?? ./dir/** – copy files that have 3 letters in the name and 2 characters in the extension to folder **dir/**

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List all `.bed` files in the directory `/home/Shared/` in a human-readable format with details.

How many AlphaFold-predicted structures (filename starts with “AF”) in cif format are there in `/home/Shared/pdbs`?

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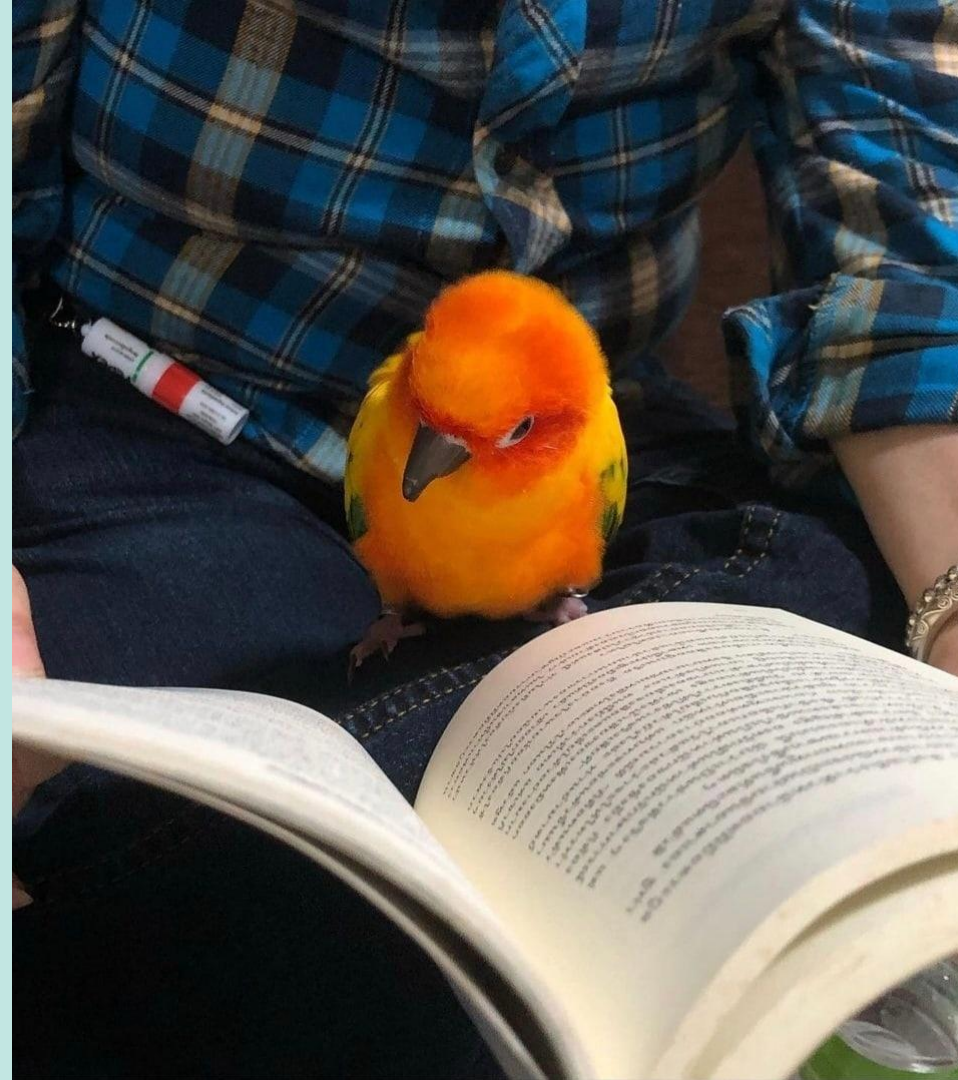
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How many AlphaFold-predicted structures (filename starts with “AF”) in cif format are there in **/home/Shared/pdbs** ?

```
ls ../Shared/pdbs/AF*.cif | wc -l
```

# Assignment



# Assignment

[https://github.com/kluwik/command\\_line/blob/main/Assignment.md](https://github.com/kluwik/command_line/blob/main/Assignment.md)

- For each task, submit a **one-line** command to Canvas (assignment will be published shortly).
- Use only **relative** paths.
- Mind the **current** directory (the current directory is stated in the assignment description).
- Deadline: **Nov 15** 12:00 (soft), Nov 22 12:00 (hard).

# Your questions

